

GOLDFISH · *CARASSIUS AURATUS*

Set up your goldfish tank the right way from day one.

A complete, checked-fact printable manual: tank sizing, filtration and cycling, diet, health triage, and the routines that keep a goldfish thriving for its full lifespan.

21 pages

Beginner to Intermediate

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't set up yet

Read pages 4 through 8 in order before buying anything. Tank size and filtration decisions are expensive to redo, and an undersized tank is the single most common goldfish mistake. Get the tank size and filtration right the first time, then use the Budget & Shopping List on page 15 to buy once.

If your goldfish is already set up

Jump to the Setup Checklist and Water Quality Targets on page 13 to audit what you have, and the Health Red Flags section starting on page 9 if anything seems off.

Print these first

- | | |
|---|---|
| ☐ Page 13: Setup Checklist & Water Quality Targets | ☐ Page 18: Daily, Weekly, Monthly & Seasonal Routine |
| ☐ Page 14: Emergency & Quick Targets Card | ☐ Page 19: Owner Log |
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WHY THIS PACKAGE EXISTS

Goldfish care sheets online are full of outdated advice: bowls, tiny tanks, "feed once a day and that's it." This package reflects current tank size, filtration, and water quality standards, cross-checked against verified sources, not recycled forum advice.

NOT A SUBSTITUTE FOR

An aquatic veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a vet call, not a page in this book.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Carassius auratus</i>
Adult size	4 to 12+ in depending on variety; fancy varieties often 6 to 8 in
Captive lifespan	10 to 15 years typical in a tank; 20 to 30+ years possible in a spacious, well-kept pond
Diet type	Omnivore; pellets plus vegetables such as peas and blanched greens
Tank minimum, fancy varieties	20 gal for one fish, plus 10 gal per additional fish
Tank minimum, single-tail varieties	55 gal for one fish (comet, common); considerably more per additional fish
Water temperature	65 to 75°F
Handling	Rarely needed; slime coat takes 24 to 48 hours to regenerate if damaged

Cost overview

Initial setup

\$150 to \$400 for the tank, filtration, substrate, decor, dechlorinator, and test kit. A heater is optional and adds \$15 to \$30 if your room temperature runs outside 65 to 75°F.

Monthly ongoing

\$10 to \$30 for food, dechlorinator, and incidental supplies like replacement filter media.

BUDGET FOR THE UNEXPECTED TOO

Aquatic vet visits, especially for conditions like dropsy, are real costs even though they're less predictable than routine expenses. Neither this package nor careful husbandry can promise you'll never need one.

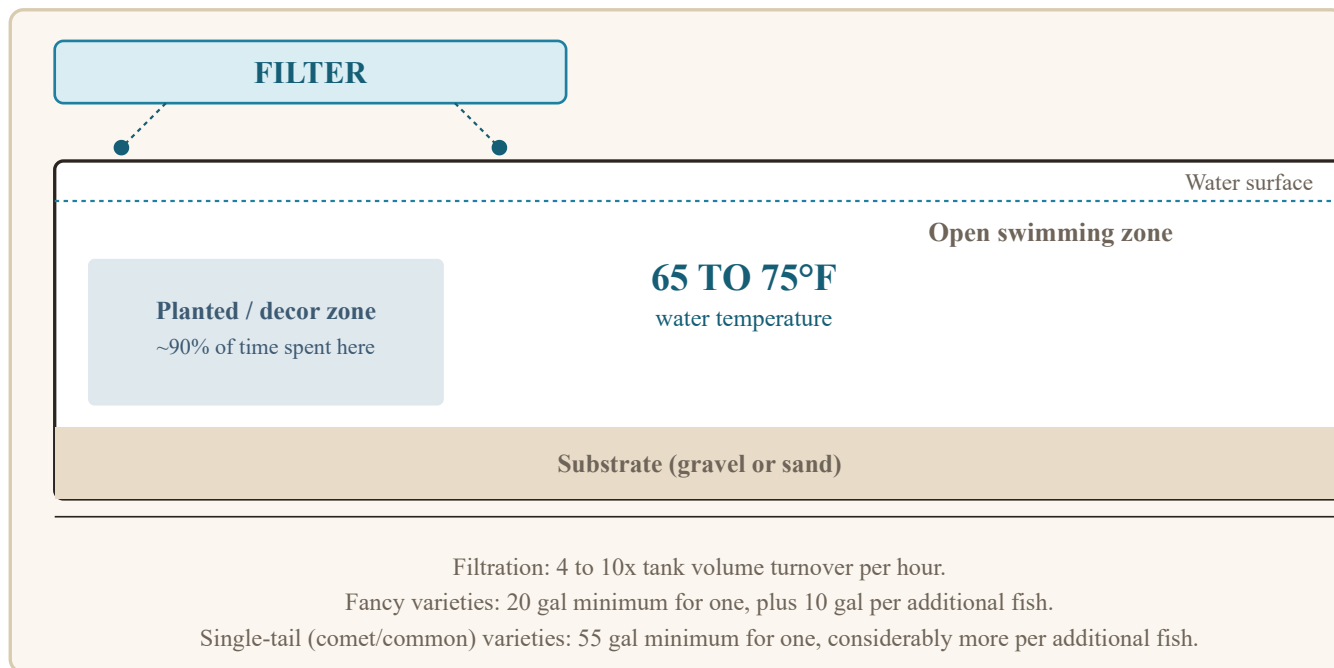
Fun facts worth knowing

- A well-kept pond goldfish can outlive many family pets, with documented individuals in the 20 to 30+ year range, versus the 10 to 15 years typical for a fish kept in a tank.
- Goldfish spend roughly 90% of their time near structure like plants and caves rather than open water, which is why layout matters more than most tank owners assume.
- The slime coat lining a goldfish's body is a living layer that regenerates in 24 to 48 hours if scraped, which is why nets and dry hands do more damage than they look like they do.

SECTION 02 · FULL CARE GUIDE

Tank size & footprint

Goldfish are one of the most undersized pets in the hobby. Fancy varieties (the rounder-bodied, shorter goldfish like orandas and ranchu) need at least 20 gallons for one fish, plus 10 gallons for each additional fish. Single-tail varieties (comets and common goldfish) grow much larger and need at least 55 gallons for one fish, with considerably more for each additional fish. Neither type is suited to a bowl or a small tank at any life stage, including as a young fish.



NEVER

Keep a goldfish in a bowl or an unfiltered container. Bowls can't hold enough water volume or filtration to keep ammonia and nitrite at safe levels, and single-tail varieties in particular will outgrow anything under 55 gallons within their first year or two.

What the diagram doesn't show

- A lid isn't required the way it is for jumping species, but it cuts evaporation and keeps debris and curious pets out.
- Leave a few inches of air space between the water surface and the lid or rim; gas exchange at the surface matters for oxygen levels.
- Plan the footprint (length and width), not just total gallons. A long, low tank offers more surface area for gas exchange than a tall, narrow one holding the same volume.

Filtration, cycling & water quality

Filtration is not optional or "nice to have," it's the system that keeps ammonia and nitrite at zero. Run filtration rated for at least 4 times your tank's total volume per hour, and aim for 5 to 10 times for a heavily stocked or heavily fed tank.

The nitrogen cycle, in practice

Fish waste and uneaten food produce ammonia. Beneficial bacteria convert that ammonia to nitrite, then convert the nitrite to nitrate. A brand-new tank has none of these bacteria yet, which is why a new setup needs to be cycled, either fishless before adding fish, or slowly with frequent water changes if fish are already in, before it can safely hold a full bioload.

TARGET	READING
Ammonia	0 ppm
Nitrite	0 ppm
Nitrate	Under roughly 20 ppm
Temperature	65 to 75°F

Water changes

Change 25 to 50% of the water weekly, more if nitrate creeps up between changes or the tank is heavily stocked. Always treat new water with a dechlorinator before adding it, and match the temperature reasonably closely to avoid shocking the fish.

TEST, DON'T GUESS

A liquid test kit for ammonia, nitrite, nitrate, and pH costs little compared to a fish lost to water you assumed was fine.

NEVER

Do a full 100% water change on an established tank except in a genuine emergency. It strips out the beneficial bacteria colony along with the water, which can restart the cycle from zero. See Cycling Failure & New-Tank Syndrome on page 11.

Diet & feeding

Goldfish are omnivores. A daily diet of pellets formulated for goldfish, supplemented with vegetables like peas and blanched leafy greens, covers their nutritional needs without the guesswork of a varied live-food rotation.

What to feed

Pellets

The base of the diet. Sinking pellets are usually a safer default than floating ones, which can lead a fish to gulp air while feeding.

Vegetables

Peas (shelled or de-shelled, thawed if frozen) and blanched greens like spinach or lettuce a few times a week add fiber and help prevent digestive issues.

How much, how often

Feed small amounts once or twice a day, only what the fish visibly eats within a couple of minutes. Uneaten food breaks down and adds directly to the ammonia load, so more food does not mean a healthier fish, it means worse water.

THE MOST COMMON MISTAKE

Overfeeding is the single most common goldfish care mistake. It drives poor water quality, which in turn drives most of the illnesses in this package far more often than any single pathogen does.

A useful habit

Keep a small container of peas or sinking pellets set aside specifically for a fish that looks off. A short fast followed by a shelled pea is the standard first response to a fish showing early swim bladder symptoms; see page 10.

Common mistakes & enrichment

Mistakes that cause the most harm

- Buying a bowl or small tank "to start," then never upgrading once the fish outgrows it. Single-tail varieties in particular need far more room than most first-time owners expect.
- Sizing the tank for the fish's current small size instead of its adult size. A goldfish can live 10 to 15 years or more, plan for that from day one.
- Undersizing filtration relative to the tank, or skipping it entirely as an "extra" expense.
- Overfeeding, and specifically feeding floating pellets that encourage gulped air.
- Skipping a fishless cycle and adding a full stock of fish to brand-new water on day one.
- Doing large emergency water changes with untreated tap water, or water at the wrong temperature.
- Handling a goldfish with dry hands or a dry net instead of guiding it into a water-filled container, which damages the slime coat.

Enrichment, by priority

Planted / structured layout

Goldfish spend roughly 90% of their time in planted or structured areas rather than open water. Live or silk plants, caves, and varied layout matter more than most owners assume.

Foraging

Scatter a portion of the pellet ration across the substrate instead of dropping it all in one spot, so the fish spends time searching rather than gulping.

Varied structure

Rearranging or adding one new cave, plant, or decoration occasionally keeps the tank visually and physically interesting without a full teardown.

Observation time

Goldfish are active, visually oriented fish that respond to routine and to their owner's presence at the tank. Regular, calm observation time doubles as your best early illness detector.

Health red flags

What to watch for, and what to actually tell the vet.

CONTACT AN AQUATIC VET PROMPTLY IF YOU SEE

Floating sideways or upside down, sinking and struggling to stay upright, a swollen belly with raised, pinecone-like scales, white spots covering the body or fins, frayed or disintegrating fins, gasping persistently at the surface, or a sudden refusal to eat with no water quality explanation.

What to tell the vet

- ☐ Current water temperature and test kit readings (ammonia, nitrite, nitrate)
- ☐ Tank size and how many fish it holds
- ☐ When symptoms started and whether they're worsening
- ☐ Date of your last water change and roughly how much

Why water quality comes up first

Most of the conditions in this section, cycling failure, new-tank syndrome, and even flare-ups of ich or fin rot, trace back to a water quality problem more often than anything else. A vet, or a good test kit, will point there before anything else for exactly that reason.

Normal vs. concerning

NORMAL	CONCERNING
Slower activity and less interest in food in cooler water	Persistent gasping at the surface or labored breathing
Brief pale color right after a big meal or a large water change	Sustained clamped fins together with lethargy
Occasional resting near the bottom or glass	A swollen belly with raised, pinecone-like scales
Mild, occasional fin nipping in a busy community tank	Frayed, discolored, or disintegrating fin edges

Swim bladder disease & ich/fin rot

Swim Bladder Disease

Cause: overfeeding, gulped air from floating pellets, constipation, or an internal abnormality affecting the swim bladder, the organ that controls buoyancy. **Signs:** floating sideways, sinking to the bottom, or visible trouble staying upright in the water. **Diagnosis:** usually made from the visible buoyancy problem itself; a vet can rule out other causes if it doesn't resolve. **Response:** fast the fish for 1 to 2 days, then offer a single shelled or de-shelled thawed pea. **Recovery:** many diet-related cases improve within days once the fish is switched to sinking pellets going forward instead of floating ones.

PREVENTION

Switch to sinking pellets by default, feed measured small portions, and keep peas on hand for the first sign of buoyancy trouble.

Ich & Fin Rot

What they are: two of the most common external diseases in freshwater fish, usually tied to poor water quality or stress rather than appearing out of nowhere. **Signs, ich:** small white spots scattered across the body and fins, resembling grains of salt. **Signs, fin rot:** frayed, ragged, or discolored fin edges that progressively shorten. **Response:** improve water quality first, test and correct ammonia, nitrite, and nitrate, then treat with an appropriate medication. Treating the disease without fixing the water it happened in tends to bring it back.

CONCRETE SIGNAL

New white spots appearing after a stressful event, such as a new fish added without quarantine or a cold snap, is common enough that it's worth testing water and checking temperature stability before assuming it's "just going around."

Cycling failure, new-tank syndrome & dropsy

Cycling Failure & New-Tank Syndrome

Cause: an ammonia or nitrite spike in a tank that isn't fully cycled yet, or one where the bacteria colony was disrupted, by a full water change, a filter media rinse in tap water, or a filter left off too long. **Signs:** gasping at the surface, red or inflamed-looking gills, lethargy, clamped fins. **Diagnosis:** a liquid test kit reading for ammonia or nitrite above 0 ppm confirms it directly. **Response:** immediate partial water changes, 25 to 50%, and retest every day or two until readings return to 0 ppm ammonia and 0 ppm nitrite. **Recovery:** fish generally recover once water quality is corrected, though gill damage from a severe or prolonged spike can leave lasting sensitivity.

WHY IT HAPPENS EVEN TO CAREFUL OWNERS

A filter is a bacteria colony as much as it is a mechanical part. Rinse media in old tank water, not tap water, and never replace all your filter media at once.

Dropsy

Signs: a swollen, bloated belly combined with scales standing out from the body in a pinecone-like pattern. **What it usually means:** organ or kidney failure, often a late-stage symptom of an underlying infection or chronic poor water quality rather than a disease on its own. **Response:** isolate the fish if you have a spare tank, and consult an aquatic vet. **Reality check:** the prognosis is generally poor by the time the pinecone scale pattern is visible, but a vet can still confirm what's actually happening and whether anything can be done.

Handling, slime coat & stress-related illness

Goldfish should rarely be handled at all. Unlike a pet that tolerates being picked up, a goldfish out of water is a fish in a medical emergency. Every unnecessary handling event is also a risk to its slime coat and its stress level.

The slime coat

A goldfish's slime coat is a living protective layer over its scales that guards against infection and parasites. It takes 24 to 48 hours to regenerate if damaged, and dry hands, a dry net dragged across the body, or rough handling are the most common ways owners damage it without realizing.

NEVER

Lift a goldfish out of water with bare hands or a dry net. If you need to move a fish, guide it into a water-filled container and move the container, not the fish.

Stress and disease are connected

A stressed fish is a fish with a suppressed immune system. Rough handling, sudden temperature swings, an under-cycled tank, or a poor diet all raise stress, and stressed fish are exactly the ones that develop ich or fin rot flare-ups from opportunistic organisms a healthy fish would otherwise shrug off. Treating the visible disease without addressing what stressed the fish in the first place tends to invite it back.

When you do need to net or move a fish

- Fill the transfer container with water from the same tank first, so temperature and chemistry match.
- Use the net to gently guide the fish into the container rather than scooping and lifting it through air.
- Keep the whole process under a few seconds of the fish being out of its home water.
- Wash your hands, with no soap residue, before any contact with tank water or decor.

Setup checklist & water quality targets

Essential setup checklist

- ☐ Tank sized correctly for your variety (20 gal fancy / 55 gal single-tail minimum)
- ☐ Filtration rated for 4 to 10x tank volume per hour
- ☐ Heater only if room temp falls outside 65 to 75°F
- ☐ Liquid test kit for ammonia, nitrite, nitrate, pH
- ☐ Dechlorinator
- ☐ Substrate (gravel or sand)
- ☐ Decor: caves, plants, varied layout
- ☐ Sinking pellets as the default food
- ☐ Water-filled container on hand for any necessary handling
- ☐ Tank fully cycled before adding a full stock of fish

Water quality targets

TARGET	RANGE
Ammonia	0 ppm
Nitrite	0 ppm
Nitrate	Under roughly 20 ppm
Temperature	65 to 75°F
Filtration turnover	4 to 10x tank volume per hour
Water changes	25 to 50% weekly

Before your goldfish comes home

- ☐ Tank fully cycled, or a plan for a slow fish-in cycle with frequent water changes
- ☐ Filtration running and turnover confirmed
- ☐ Temperature stable and confirmed with a thermometer, not by feel
- ☐ Food, dechlorinator, and test kit on hand before day one

Emergency & quick targets

CALL AN AQUATIC VET NOW IF

Swollen belly with pinecone-like raised scales · gasping persistently at the surface · floating sideways or sinking and unable to right itself · white spots covering body/fins with rapid spread · frayed or disintegrating fins · sudden refusal to eat with no water quality explanation

READING	TARGET
Ammonia	0 ppm
Nitrite	0 ppm
Nitrate	Under roughly 20 ppm
Temperature	65 to 75°F
Filtration turnover	4 to 10x tank volume per hour
Water change	25 to 50% weekly

HAVE THIS READY WHEN YOU CALL

Ammonia, nitrite, and nitrate readings, water temperature, tank size and fish count, last water change date, and diet over the last two weeks.

Budget & shopping list

ITEM	TYPICAL COST
Tank	\$50 to \$150
Filtration (per 10 gal)	\$5 to \$15
Heater (optional)	\$15 to \$30
Substrate, decor, dechlorinator & test kit	\$30 to \$60
Total initial setup	\$150 to \$400
Monthly ongoing (food, dechlorinator, supplies)	\$10 to \$30

PLAN FOR, DON'T COUNT ON AVOIDING

Aquatic vet visits, especially for conditions like dropsy, are real costs even though they're less predictable than routine expenses. A small emergency fund matters as much as any item on this list.

WHERE NOT TO CUT CORNERS

Tank size and filtration are the two line items most directly tied to the health conditions in this package. Save on decor and extras before you save here.

First 30 days checklist

Week 1

- ☐ Confirm temperature, and run a new tank empty of fish for 24 to 48 hours while checking that filtration and heating hold steady
- ☐ Add fish gradually rather than fully stocking on day one
- ☐ Test water daily for the first two weeks

Week 2 to 3

- ☐ Test ammonia, nitrite, and nitrate every few days and log the readings
- ☐ Establish a consistent feeding routine: small amounts, once or twice a day
- ☐ Start the Owner Log on page 19

Week 4

- ☐ Review the ammonia, nitrite, and nitrate trend in the Owner Log; readings should be settling toward 0/0/under 20
- ☐ Set a recurring reminder for weekly 25 to 50% water changes
- ☐ Introduce first enrichment items from page 20

Days 31 to 90

- ☐ Confirm the tank is fully cycled: consistent 0 ppm ammonia and 0 ppm nitrite between water changes
- ☐ Check filter media and test kit reagents against the Equipment Log on page 19
- ☐ Look back at the Owner Log for any slow drift in water readings or appetite
- ☐ Start rotating enrichment on the monthly schedule from page 20

Symptom quick reference

SYMPTOM	LIKELY CAUSE	ACTION
Floating sideways or sinking	Swim bladder disease	Fast 1 to 2 days, then a pea (page 10)
White spots on body/fins	Ich	Improve water quality, treat (page 10)
Frayed or discolored fin edges	Fin rot	Improve water quality, treat (page 10)
Swollen belly, pinecone-like scales	Dropsy	Isolate, vet promptly (page 11)
Ammonia or nitrite reading above 0 ppm	Cycling failure / new-tank syndrome	Partial water change, retest (page 11)
Gasping persistently at the surface	Poor water quality or oxygen	Test water, water change immediately (page 11)
Torn fins or cloudy patches after handling	Slime coat damage	Minimize handling, watch for infection (page 12)
Loss of appetite, no water quality cause found	Several possible causes	Test water and temp first, vet if it persists
Cloudy water despite filtration	Overfeeding or incomplete cycle	Test water, reduce feeding, check filter (page 6)
Sudden change between log entries	Several possible causes	Vet visit, bring the Owner Log (page 19)

Daily, weekly, monthly & seasonal routine

Daily

- ☐ Check water temperature
- ☐ Feed per the schedule on page 7
- ☐ Quick visual check: swimming behavior, fins, appetite, color

Weekly

- ☐ Test ammonia, nitrite, nitrate, and pH
- ☐ Change 25 to 50% of the water
- ☐ Log readings in the Owner Log on page 19

Monthly

- ☐ Rinse filter media in old tank water, never tap water
- ☐ Check test kit reagents against their expiry date
- ☐ Reassess whether decor and layout still suit your fish's current size

Seasonal

- ☐ Watch for room temperature swings between summer and winter, especially without a heater
- ☐ Recheck filtration turnover if you've added fish since setup

Owner log

Photocopy this page, or track digitally. Consistency matters more than the format.

DATE	WATER °F	AMMONIA	NITRITE	NITRATE	APPETITE	NOTES

WHY BOTHER LOGGING

A slow drift in water readings is the single best early warning system you have. Most of the conditions in this package show up in test results before they show up as visible symptoms.

Equipment log

ITEM	LAST CHECKED/CHANGED	NEXT DUE
Filter media		
Test kit reagents (expiry)		
Heater function check		

Vet visit log

DATE	REASON	FOLLOW-UP

Enrichment checklist

Setup

- ☐ Live or silk plants and caves covering a meaningful portion of the tank footprint
- ☐ Varied layout rather than a single open rectangle of water
- ☐ At least one cave or hide-style structure

Monthly review

- ☐ Rotate or reposition one piece of decor to refresh the layout
- ☐ Scatter-feed a portion of the pellet ration across the substrate at least once
- ☐ Reassess whether decor still suits your fish's current size

Enrichment log

DATE	ACTIVITY	HOW IT WENT

Glossary

Cycling	The process of establishing a colony of beneficial bacteria that converts ammonia to nitrite, then nitrite to nitrate, in a new or disrupted tank.
Ammonia	A toxic waste product from fish waste and uneaten food; the target reading is 0 ppm in a healthy, cycled tank.
Nitrite	A byproduct of ammonia breakdown, also toxic to fish; the target reading is 0 ppm.
Nitrate	The end product of the nitrogen cycle, far less toxic than ammonia or nitrite but still harmful at high levels; keep it under roughly 20 ppm with regular water changes.
Swim bladder	An internal organ that controls a fish's buoyancy; problems with it cause floating, sinking, or trouble staying upright.
Dropsy	A symptom complex, a swollen belly and pinecone-like raised scales, usually indicating organ or kidney failure.
Slime coat	The living protective layer over a goldfish's scales; it takes 24 to 48 hours to regenerate if damaged.
Turnover	How many times per hour a filter cycles the tank's total water volume; aim for 4 to 10 times per hour.
Dechlorinator	A water conditioner that neutralizes chlorine and chloramine from tap water before it reaches the fish.

Compiled from aquarium-keeping and veterinary sources, cross-checked against current Beastly Facts guides at time of publication. This package is a husbandry reference, not a substitute for an aquatic veterinarian, and is not a live website mirror. No external links or website CTAs appear in this document.

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